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Detection of *Rickettsia* spp. in Fleas Collected from Small Mammals in Slovakia, Central Europe

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Abstract

This survey is aimed at investigation of species composition of fleas removed from small terrestrial mammals captured in rural, suburban, and urban types of habitat and molecular screening of the presence and diversity of *Rickettsia* species in collected ectoparasites. In total, 279 fleas (Siphonaptera) belonging to 9 species of 2 families, Ceratophyllidae and Hystrichopsyllidae, were collected from 115 (46%) out of 250 trapped small mammals of eight species (*Apodemus agrarius*, *Apodemus flavicollis*, *Apodemus uralensis*, *Myodes glareolus*, *Microtus arvalis*, *Microtus subterraneus*, *Crocidura leucodon*, and *Sorex minutus*). *Rickettsia* spp. were found in 2.5% (7/279) of tested fleas, namely in *Ctenophthalmus agyrtes*, *Ctenophthalmus solutus*, *Ctenophthalmus uncinatus*, *Megabothris turbidus*, and *Amalareus penicilliger*. *Rickettsia felis*, *Rickettsia helvetica*, and unidentified *Rickettsia* species were detected in fleas infesting small mammals in Eastern Slovakia. The results of the study suggest that some species of rickettsiae have a different range of arthropod vectors.

Keywords: fleas, *Rickettsia felis*, *Rickettsia helvetica*, small mammals, Slovakia

Introduction

Rickettsiae are obligate intracellular Gram-negative bacteria from the order Rickettsiales with a worldwide distribution. Many species are responsible for serious diseases in humans and other mammals (Raoult and Roux 1997, Parola et al. 2013). They are transmitted through arthropod vectors such as ticks, mites, fleas, and lice (Raoult and Roux 1997, Weinert et al. 2009, Miťková et al. 2015). Recent studies have shown that genus *Rickettsia* could be divided into several different phylogenetic groups: the spotted fever group (SFG) (*e.g.*, *Rickettsia africae*, *Rickettsia sibirica*, and *Rickettsia slovaca*); the typhus group (TG) (*Rickettsia typhi* and *Rickettsia prowazekii*); the transitional group (TRG) (*Rickettsia akari* and *Rickettsia felis*); and as own groups *Rickettsia helvetica*, *R. canadensis*, and *Rickettsia bellii* (Murray et al. 2016).

In Slovakia, six species of rickettsiae have thus far been identified in vectors (*R. slovaca*, *Rickettsia raoultii*, *Rickettsia monacensis* strains IRS3 and IRS4, *R. helvetica*, *R. africae*, and *R. felis*) (Sekeyová et al. 2012, 2013, Špitalská et al. 2015). In this region, *R. africae*, *R. felis*, and *R. helvetica* have been detected in fleas. *R. africae*, the etiological agent of African tick-bite fever, was identified in *Ceratophyllus garei* flea collected from migrating bird species, *Acrocephalus scirpaceus*, during spring migration from Africa to Europe (Sekeyová et al. 2012). In the world, *R. africae* has also been detected in several species of ticks (*Rhipicephalus*, *Hyalomma*, *Haemaphysalis*, or *Amblyomma*) (Mediannikov et al. 2012, Orkun et al. 2014).

In Slovakia, *R. felis* was detected in *Ctenophthalmus solutus* flea collected from *Apodemus agrarius* (Špitalská et al. 2015). *R. felis*, the causative agent of flea-borne spotted fever, also known as cat flea typhus, is primarily associated with the cat flea *Ctenocephalides felis*, which is the main arthropod reservoir as well as the vector (Abdad et al. 2011). *R. felis* has also been identified molecularly in other species of fleas, ticks, mites, lice, and mosquitoes (*Aedes* sp.) (Socolovschi et al. 2012, Brown and Macaluso 2016). Members of the *genus Rickettsia* are primarily transmitted vertically, by transstadial and transovarial transmission, within a flea population. However, horizontal transmission of *R. felis* between cat fleas was also documented (Wedincamp and Foil 2002, Hirunkanokpun et al. 2011).

Fleas are holometabolous insects with hematophagous images. They are significant ectoparasites of many mammals and birds but are mainly associated with wild small mammals (Krasnov 2008). In previous surveys in Slovakia, >30 species of fleas infesting small mammals were recorded (Rosický

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1957, Jurík 1962, Stanko 1987, 1992). Fleas are known vectors of various pathogens. including the etiological agents of plague (*Yersinia pestis*), bartonellosis (*Bartonella henselae*), and rickettsiosis (*R. typhi*, *R. prowazekii*, *R. africae*, and *R. felis*) (Raoult and Roux 1997, Bitman et al. 2010). The host becomes usually infected by flea feces coming into contact with injured skin (Azad and Beard 1998).

Recently, several studies regarding the fleas and rickettsiae they transmit were published in Slovakia (Sekeyová et al. 2012, Špitalská et al. 2015, Víchová et al. 2018). However, information on the diversity of rickettsiae in fleas in different regions of Slovakia is still limited. Therefore, the main aim of this study was to screen fleas collected from small mammals that were captured in Eastern Slovakia (Central Europe), for the presence and diversity of *Rickettsia* spp. In this publication, examinations of fleas represent only one part of the complex results focused on the role of arthropods parasitic on small mammals in the circulation of rickettsia.

Material and Methods

Fleas sampling

Fleas were collected from rodents and shrews at three sampling sites in Eastern Slovakia during the years 2014–2016. The sampling sites were described in detail in Heglasová et al. (2018). The first study site, a Botanical garden, with deciduous forest vegetation, is located in the center of Košice city (208 meters a.s.l.; 48°446 · 84″N; 21°14′16 · 14″E). It represents an urban habitat with a significant anthropogenic impact. The second site, Čermel' valley, is characterized by the presence of mixed forest vegetation, with a predominance of beech, hornbeam, and spruce (208–600 meters a.s.l.; 48°45′46 · 67″N; 21°8′8 · 17″E), and it represents a suburban habitat. The third site, with mixed forest vegetation and a predominance of beech, oak, and hornbeam, is located in the Slovak Karst National Park, near the Hrhov village (200–220 meters a.s.l.; 48°34′53.9″N; 20°46′44.4″E), and it represents a rural habitat.

Live-trapping of small mammals was carried (two times per year) out during May/June and September/October in years from 2014 to 2016, respectively. Rodent sampling was described in detail in Heglasová et al. (2018). Captured small mammals were euthanized in accordance with the licenses of the Ministry of Environment of the Slovak Republic No. 4874/2011–2.2 and 4559/2015–2.3. Each rodent individual was checked for the presence of fleas. All collected fleas were stored in 70% ethanol until identification. The species and sex of fleas were determined based on morphological characteristics under a light microscope using the morphological keys and published descriptions (Rosický 1957).

DNA extraction and molecular analyses

Fleas were washed with sterile water, dried on air, transferred to individual tubes, and crushed with a sterile scalpel. Genomic DNA was extracted individually from each flea using alkaline hydrolysis method (Rijpkema et al. 1996). All DNA samples were stored at −20°C for further processing.

Subsequently, all DNA samples were screened for the presence of *Rickettsia* spp. by nested PCR amplification of the 381-bp long fragment of the citrate synthase gene (*gltA*), using the genus-specific primers RpCS.877p and RpCS.1258n (Regnery et al. 1991) and inner primer pair RpCS.896p and RpCS.1233n (Choi et al. 2005) resulting in 337-bp long PCR product. To further characterize *Rickettsia* species, chosen three out of seven *gltA* positive amplicons were subsequently analyzed for the presence of the fragments of another gene loci (*ompA* and *ompB*) encoding outer membrane proteins A and B (Regnery et al. 1991, Roux et al. 1996, Roux and Raoult 2000), with primers RR190.70F/RR190.701R and 120–2788/120–3599, respectively.

R. helvetica identification was based on the PCR amplification of the *gltA* and *ompB* genes, and *R. felis* was identified based on amplification of the *gltA*, *ompA*, and *ompB* genes (Roux et al. 1996, Roux and Raoult 2000, Portillo et al. 2017). After PCR analyses, all positive samples with amplified *gltA*, *ompA*, and *ompB* gene fragments were purified using Isolate II PCR and Gel Kit (Bioline) and sequenced.

Accession numbers of obtained nucleotide sequences

Nucleotide sequences of the fragments of *gltA* gene obtained in this study from fleas were submitted to the GenBank database under the following accession numbers: *R. felis* (MN276063), *R. helvetica* (MN276064, MN276065).

Results

Flea communities

In total, 279 fleas (Siphonaptera) belonging to nine species of two families, Ceratophyllidae and Hystrichopsyllidae, were screened for *Rickettsia* spp. The most frequent flea species was *Ctenophthalmus agyrtes* ($n=105$), followed by *C. solutus* ($n=63$), *Ctenophthalmus assimilis* ($n=49$), *Megabothris turbidus* ($n=17$), *Amalareus penicilliger* ($n=15$), *Hystrichopsylla orientalis* ($n=15$), *Ctenophthalmus uncinatus* ($n=12$), *Doratopsylla dasycnema* ($n=2$), and *Nosopsyllus fasciatus* ($n=1$). The greatest abundance and diversity of the fleas removed from small mammals were detected in rural habitat of Slovak Karst region with totally collected 169 fleas of 8 species, followed by suburban habitat, Čermel', with 97 fleas of 4 species, and urban area of Botanical Garden in the Košice city, with 13 fleas of 2 species. Two most common flea species (*C. agyrtes*, *C. solutus*) were detected in all surveyed regions (Table 1).

All 250 small mammals trapped in the previous study by Heglasová et al. (2018) were screened for the presence of fleas in this study. Totally, fleas were removed from 115 (46%) out of 250 trapped small mammals belonging to eight species (*A. agrarius*, *Apodemus flavicollis*, *Apodemus uralensis*, *Myodes glareolus*, *Microtus arvalis*, *Microtus subterraneus*, *Crocidura leucodon*, and *Sorex minutus*).

The presence of Rickettsia *spp. in fleas*

The presence of rickettsiae was identified in *C. agyrtes*, *C. solutus*, *C. uncinatus*, *M. turbidus*, and *A. penicilliger* fleas collected from *A. flavicollis*, *A. agrarius*, and *M. glareolus*, respectively. Fleas infesting small mammals tested negative for SFG rickettsiae in the previous study realized in this region (Heglasová et al. 2018). The overall prevalence of *Rickettsia* spp. in the tested fleas was 2.5% (7/279). Five infected fleas originated from Čermel', one flea from Botanical garden, and one from the Slovak Karst region.

All seven *gltA*-PCR positive amplicons were sequenced. Nucleotide sequence of *gltA* gene (MN276065), amplified

Table 1. Examined Fleas (Siphonaptera) from Small Mammals in Three Different Sites of East Slovakia

Flea species	*Slovak Karst (natural)*	*Čermel' (suburban)*	*Botanical garden (urban)*	*Hosts of fleas*
Amalareus penicilliger	–	15 (9♀, 6♂)	–	*Apodemus flavicollis* *Myodes glareolus*
Ctenophthalmus agyrtes	49 (32♀, 17♂)	54 (32♀, 22♂)	2♀	*Apodemus agrarius* *A. flavicollis* *M. glareolus* *Microtus arvalis* *Microtus subterraneus*
Ctenophthalmus assimilis	49 (27♀,22♂,)	–	–	*A. agrarius* *A. flavicollis* *M. arvalis*
Ctenophthalmus solutus	35 (19♀, 16♂)	17 (8♀, 9♂)	11(7♀, 4♂)	*A. agrarius* *A. flavicollis* *M. arvalis* *M. glareolus*
Ctenophthalmus uncinatus	1♀	11 (4♀, 7♂)	–	*A. flavicollis* *M. glareolus*
Doratopsylla dasycnema	2♂	–	–	*Crocidura leucodon* *Sorex minutus*
Hystrichopsylla orientalis	15 (7♀, 8♂)	–	–	*A. agrarius* *A. flavicollis* *C. leucodon* *M. arvalis*
Megabothris turbidus	17 (11♀, 6♂)	–	–	*A. agrarius* *A. flavicollis* *M. glareolus*
Nosopsyllus fasciatus	1♂	–	–	*A. flavicollis*
Total	169 (97♀, 72♂)	97 (53♀, 44♂)	13 (9♀, 4♂)	

from *C. agyrtes* male flea, collected from rickettsia-negative *A. agrarius* from Čermel', showed 100% identity with at least 50 *R. helvetica* isolates (MK875721-MK875725, MK875700-MK875719) from *Ixodes* spp. ticks from Slovakia that are deposited in the Genbank database. Nucleotide sequence of *gltA* gene (MN276064), amplified from *C. uncinatus* female flea collected from rickettsia-negative *A. flavicollis* from the locality Čermel' showed 99.69% similarity with *R. helvetica* (MG190375) from small mammal's ear biopsy from Slovakia and 99.08% similarity with *R. helvetica* (MF673863) from *Ixodes ricinus* tick from Slovakia. The fragments of the *ompB* gene were not successfully amplified in tested samples of *R. helvetica*.

The third partial nucleotide sequence of *gltA* gene (MN276063), amplified from *C. solutus* female flea collected from rickettsia-negative *A. agrarius* from Botanical Garden in Košice city, showed 99.08% similarity with *R. felis* (JF448471-JF448473) from South Korean *A. agrarius* mice. The *ompA* gene fragment sequences from *C. solutus* showed 98.08% similarity with *R. felis* (AF191026) and 98.07% similarity with *R. felis* (HM636635, EU012496) from infected fleas from dogs in Yucatan and Mexico, respectively. The nucleotide sequence of the *ompB* gene fragment sequences showed 99.87% similarity with *R. felis* (MG266432) from *Arvicola amphibius* rodent in Germany.

Four out of seven *gltA*-PCR positive amplicons were not closely identified. Three partial nucleotide sequence of *gltA* gene amplified from two male and one female of *A. penicilliger* collected from *M. glareolus* captured in Čermel' showed 96.83% similarity with uncultured *Rickettsia* sp. clone 42-2 (MF459640) from *C. felis* in Uganda and 96.78% similarity with uncultured *Rickettsia* sp. isolate Turbo (MK860201) from *Homo sapiens* in Colombia. One partial nucleotide sequence of *gltA* gene amplified from one *M. turbidus* male from *M. glareolus* captured in Hrhov, Slovak Karst region showed 99.66% similarity with uncultured *Rickettsia* sp. clone (KX179467) from *M. turbidus* fleas collected from *A. agrarius* in Slovakia.

Discussion

In this study, *R. felis*, *R. helvetica*, and *Rickettsia* sp. were detected in fleas removed from wild small mammals captured in three different habitat types in Eastern Slovakia. The overall prevalence of rickettsiae was 2.5% (7/279). The low prevalence of *Rickettsia* spp. indicates that fleas seem to be less important in the transmission of *Rickettsia* species than ticks. Small mammals were analyzed for the presence of rickettsiae in the previous study of Heglasová et al. (2018) with prevalent 11% (27 positive/245 tested rodents).

In total 250 small mammals of 11 species *A. agrarius*, *A. flavicollis*, *A. uralensis*, *Micromys minutus* (Muriade), *M. glareolus*, *M. arvalis*, *M. subterraneus* (Cricetidae) and *C. leucodon*, *Crocidura suaveolens*, *Neomys fodiens*, and *S. minutus* (Soricidae) were captured in rural, suburban, and urban habitats of Eastern Slovakia. Ear biopsies of 245 individuals, belonging to eight species, were examined for the presence of SFG rickettsiae in the study previously published by Heglasová et al. (2018). All seven *Rickettsia*-positive fleas from this study were collected from *Rickettsia*-negative rodents.

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Different degrees of host specificity were observed in individual flea species. Most of the fleas collected in this study have a broad host range and infest wild mice, voles, and insectivores. *Rickettsia*-positive flea species from this study belong into generalists (*A. penicilliger*, *C. agyrtes*, and *M. turbidus*) and species with narrower host range (*C. solutus* and *C. uncinatus*) (Rosický 1957). Previous surveys showed a similar composition of flea communities in small mammals.

R. felis was identified in *C. solutus* female flea collected from *A. agrarius* inhabiting the urban area of Košice city. Similar findings were published by Špitalská et al. (2015); *R. felis* was identified in *C. solutus* male flea collected from *A. agrarius*. *R. felis* circulates through a broader spectrum of vectors and has been detected in at least 30 species of fleas, ticks, mites, and lice (Abdad et al. 2011, Brown and Macaluso 2016, Thu et al. 2019). This species of rickettsia is primarily associated with the cat flea *C. felis*, which is the main arthropod reservoir as well as the vector (Abdad et al. 2011), but it was also identified in other species of fleas from the genera *Xenopsylla*, *Archeopsylla*, *Anomiopsyllus*, *Hystrichopsylla*, and *Ctenophthalmus* removed from rodents and insectivores (Stevenson et al. 2005, Bitam et al. 2006, Radzijevskaja et al. 2018). This rickettsia species have been identified in ticks from genera *Rhipicephalus*, *Ornithodoros*, and in mites from the genera *Laelaps* and *Hyperlaelaps* (Reeves et al. 2006, Oliveira et al. 2008, Parola et al. 2013, Radzijevskaja et al. 2018).

In this study, *R. helvetica* was identified in *C. agyrtes* males collected from *A. agrarius* and *C. uncinatus* female from *A. flavicollis*. Similarly, Špitalská et al. (2015) identified *R. helvetica* in *C. agyrtes* males collected from *A. agrarius*. In the study of Sprong et al. (2009), *R. helvetica* was identified in fleas amplified from *M. glareolus* and *Apodemus sylvaticus* in the Netherlands. *I. ricinus* tick is the main vector of the most common species *R. helvetica* across Europe (Oteo and Portillo 2012). *R. helvetica* has been identified in >10 tick species of genera *Ixodes*, *Haemaphysalis*, and *Dermacentor* (Parola et al. 2005, Hornok et al. 2010, Tijsse-Klasen et al. 2013, Mărcuțan et al. 2016). In Slovakia, the presence of *R. helvetica* and so *R. monacensis* was also identified in ectoparasitic mites Laelapidae and Trombiculidae infesting rodents (Miťková et al. 2015).

Research and study of rickettsiae have revealed the great diversity of the hosts involved in their circulation. Species of rickettsiae such as *R. felis*, *R. helvetica*, *R. africae*, or *R. monacensis* are associated with several different species of ticks belonging to different genera or subfamilies, as well as with fleas or mites. These species belong to a group of rickettsiae that seem to have less specific preferences on arthropod vectors (Heglasová et al. unpublished data).

In contrast, there is a group of rickettsiae that seem to be associated with several tick species within the same genus. For instance, *R. slovaca* associated with several tick species within the same genus, mainly with *D. marginatus* and *D. reticulatus* across Europe (Sekeyová et al. 1998, Parola et al. 2005, 2013, Špitalská et al. 2012, Minichová et al. 2017). Similar findings were published by Socolovschi et al. (2009) and Merhej and Raoult (2011). The results of the study suggest that some species of rickettsiae have a different range of arthropod vectors.

Conclusions

Altogether, nine flea species were recorded on sampled small mammals from Slovakia. The occurrence of human pathogenic *R. felis* and *R. helvetica* was recorded in fleas removed from small mammals collected in habitats with various degree of anthropogenic use (urban, suburban, and rural).

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